



Supplementary materials for

Liyuan CHEN, Gaoguo JIA, Dongsheng GU, Jiangpeng YAN, Yuhang JIANG, Xiu LI, Xiaojun ZENG, 2025. MENTOR: a multi-agent framework for event and narrative trend prediction with optimized reasoning. *Front Inform Technol Electron Eng*, 26(10):1847-1861. <https://doi.org/10.1631/FITEE.2500608>

1 Evaluation for trending event detection

The following are the definitions and calculation details of the evaluation metrics for trending event detection:

$$\left. \begin{aligned} \text{ERR} &= \frac{N_{\text{mentions}}}{N_{\text{total_posts}}} \\ \text{ICR} &= \frac{N_{\text{reporting_influencers}}}{N_{\text{total_influencers}}} \\ \text{KPQ} &= \frac{N_{\text{key_points}}}{N_{\text{total_points}}} \\ \text{SDI} &= \frac{1}{N} \sum_{i=1}^N (X_i - \mu)^2 \end{aligned} \right\} E = - \sum_{i=1}^n p_i \log_b(p_i) \quad (\text{S1})$$

where N_{mentions} represents the number of times a specific event is mentioned across all public accounts, and $N_{\text{total_posts}}$ denotes the total number of posts published by all public accounts within the same time frame. $N_{\text{reporting_influencers}}$ represents the number of influential public accounts reporting on a specific event, and $N_{\text{total_influencers}}$ denotes the total number of influential public accounts. $N_{\text{key_points}}$ represents the number of associated key points extracted from the text, and $N_{\text{total_points}}$ denotes the total number of key points in the text related to the specific event. N represents the total number of key points, X_i denotes the sentiment score of the i -th key point, and μ is the mean sentiment score of all key points. E represents the current entropy value, which is derived from the computation of four heat metrics.

2 Statistical significance of performance gains

We assess the significance of Mentor's improvement over the strongest baseline (SEP_O1) using the Wilcoxon signed-rank test on 30 bootstrap samples of industry rankings (Spearman correlation). Results are shown in Table S1.

Table S1 Wilcoxon test results ($n = 30$)

Market	Statistic (W)	p -value
A-share	29.0	3.24×10^{-6}
S&P 500	122.0	0.022

Both improvements are statistically significant ($p < 0.05$). The much stronger significance on A-share is likely due to its broader time coverage and larger sample size, which yield more stable estimates. The statistic W is the smaller of the signed rank sums; lower values indicate more consistent outperformance.

3 Prompt templates

We provide representative prompt templates used in the MENTOR framework. For a given task, multiple prompt templates may be utilized for two primary reasons. First, some templates are deliberately designed for ablation studies. Second, the outputs produced by certain prompts on specific models may deviate from the required format specifications. To enhance robustness and ensure compatibility, the system is configured to sequentially attempt the corresponding prompt templates in a predefined order.

3.1 Hot event summarization

Based on the following stock market-related hot events, please summarize the top 10 hot events that occurred, ranked by their impact on the stock market: hot_events

3.2 Hot event prediction

Based on the following stock market-related hot events from last week, predict the top 10 hot events that will occur in the next week. Write in summary format rather than prediction format, ranked by their impact on the stock market.

Last week events: last_week_events

Improvement guidance: solution

3.3 Improvement suggestions templates

Template 1: Original

The following are hot events that occurred in a certain week: actual_events

Below is a previous prediction by a certain large model for events that might occur in that week: prediction

Please score this model's prediction out of 10 points, adding one point for each correct prediction, and provide potential improvement ideas. Do not consider adding new quantitative models or data, but rather correct wrong thinking patterns and provide correct thinking approaches to avoid overfitting to single data instances. The final improvement ideas will become prompts for new models. Please enclose them in square brackets [], with only one set of brackets containing all prompts.

Template 2: Direct

Compare the predicted events with actual events and provide improvement suggestions:

Actual events: actual_events

Predicted events: prediction

Please provide improvement suggestions for future predictions. Format your final suggestions as: [your suggestions here]

Template 3: Simplified

Based on the prediction accuracy, provide improvement guidance for better future predictions.

Actual: actual_events

Predicted: prediction

Improvement suggestions: [insert suggestions here]

Template 4: Analytical

Performance Analysis:

Expected outcomes: actual_events

Model predictions: prediction

Please analyze the differences and provide methodology improvements. Format: [analytical recommendations]

Template 5: Research feedback

Evaluation Summary:

Observed events: actual_events

Forecasted events: prediction

Assess accuracy and suggest enhancement strategies. Use format: [strategy improvements]

3.4 Industry ranking prediction templates

Template 1: Direct hot_events

The above are market events related to market_desc from a recent week. Below are the weekly performance data of industry_count market_desc industries: result_str

Based on this information, please analyze and provide a ranking of industry performance for the upcoming week. Output format: ['Industry1', 'Industry2', 'Industry3', ... 'Industryindustry_count'].

additional_prompt

Template 2: Analytical

Market Analysis Context:

Recent market developments: hot_events

Historical performance data for industry_count market_desc industries: result_str

Task: Analyze the trends and provide a performance ranking for next week's market outlook. Please list industries in order of expected performance. Format: Python list with industry names.

additional_prompt

Template 3: Academic

Financial Market Research:

Market events summary: hot_events

Industry performance metrics for market_desc market (industry_count sectors): result_str

Please conduct a trend analysis and suggest the likely performance ranking for the following period. Return as a list format: ['Sector1', 'Sector2', etc.].

additional_prompt

Template 4: Data-driven

Data Analysis Request:

Weekly market data for industry_count industries: result_str

Recent market context: hot_events

Based on the data patterns, please arrange the industries in order of expected performance. Output: list format.

additional_prompt

3.5 Ranking evaluation templates

Template 1: Professional Prediction Analysis:

Context: prediction_context

Predicted ranking: predicted_ranking

Actual performance data: actual_str

Please evaluate the prediction accuracy on a scale of 1-10 and suggest methodological improvements. Format suggestions as: [improvement ideas]

Template 2: Academic review

Research Evaluation:

Methodology context: prediction_context

Forecast results: predicted_ranking

Observed outcomes: actual_str

Provide an assessment score (1-10) and recommendations for methodology enhancement. Enclose recommendations in brackets: [suggestions]

Template 3: Analytical feedback

Performance Review:

Analysis background: prediction_context

Projected outcomes: predicted_ranking

Actual market results: actual_str

Rate the forecasting accuracy (1-10) and offer analytical improvements. Format: [enhancement recommendations]

Template 4: Simple comparison

Comparison Analysis:

Prediction: predicted_ranking

Reality: actual_str

Context: prediction_context

Score the accuracy (1-10) and suggest better approaches. Use format: [improvement suggestions]

3.6 Extracting score templates

Template 1: Original

Please extract the prediction score given by the large model from the following text. Extract the total combined score, the score obtained in a ten-point scale, do not include /10, only extract numbers, enclose in[]: text

Template 2: Direct

Extract the numerical score (0-10) from this text: text

Score: [your answer]

Template 3: Simple

What is the score mentioned in this text? Format: [score]

Text: text

3.7 Updating solution with feedback templates

Template 1: Original

current_solution

The above is the original prompt (if any). The following is a supplement to the original prompt:feedback

Please generate a new prompt based on the supplement. Format your response as: [new prompt here]

Template 2: Direct

Current guidance: current_solution

New feedback: feedback

Please combine them into an improved guidance. Provide your response as: [improved guidance]

Template 3: Simple

Improve this guidance: current_solution

Using this feedback: feedback

New guidance: [your response]

4 Data samples of Chinese KOL articles texts and English financial news texts

4.1 Data samples of Chinese KOL articles texts

Overview

This part provides data samples of English news texts, which contain copyrighted content licensed from Reuters. Due to the licensing agreement, the complete dataset cannot be made publicly available.

Data structure

The data is organized in a tabular format with the following columns:

ID: A unique identifier for each text entry.

TITLE: The title of the relevant content, indicating the text's topic.

AUTHOR: The name of the author or the organization behind the text.

CONTENT_TIME: The date and time that the content was created.

URL: The Web link to the original content, which has been partially masked for data protection.

CONTENT: The actual text content from which potentially identifying information has been removed.

Sample entries

Here is a sample entry demonstrating the above - mentioned data structure:

ID: [Masked ID 1]

TITLE: Revival of Weimar by Baoneng Group, Aiming for 120 Billion in 5 Years

AUTHOR: 21st Century Business Review

CONTENT_TIME: 2025 - 09 - 09 19:52:10

URL: [Masked URL 1]

CONTENT: Weimar Automobile is seeing a revival opportunity. On September 6th, it and Baoneng Automobile jointly issued a "White Paper to Suppliers", stating that they are fully promoting the resumption of mass production at the Wenzhou base and ensuring the production of over 10,000 EX5 and E.5 models by the end of the year. The new shareholder, Xiangfei Automobile, which has close ties to Baoneng, plans an initial investment of 1 billion yuan for factory equipment upgrades, supply chain restoration, and product development. It aims to produce 100,000 vehicles by 2026 with an expected revenue of nearly 10 billion yuan, and challenges a production target of 1 million vehicles and a revenue of 120 billion yuan by 2030. However, Weimar has debt issues. The ordinary creditors of the four restructured Weimar companies with claims of 150,000 yuan or less can receive full - cash settlement within 6 months. Those with larger claims can get 150,000 yuan in cash, and the remaining amount will be included in a trust for later proportional settlement. The article also discusses the company's production plans, market situation, and recruitment efforts.

4.2 Data samples of English financial news texts

Overview

This part provides data samples of English news texts, which contain copyrighted content licensed from Reuters. Due to the licensing agreement, the complete dataset cannot be made publicly available.

Data structure

The data is presented in a tabular format with the following columns:

md5_doc_id: A unique identifier for each news document in MD5 hash format.

headline: The title or headline of the news article, summarizing the main topic.

content: The full text content of the news piece, with detailed reports related to the headline.

doc_url: The web link to the original news document, which has been partially masked.

pub_code: A code associated with the publication.

source: The source of the news.

section: The section of the publication where the news appears.

author: The author of the news article.

region: The region related to the news, providing geographical context.

Sample entries

Here is a sample entry demonstrating the above - mentioned data structure:

md5_doc_id: [Masked ID 2]

headline: Carbon Credit Trading Platform Market Estimation Worth \$556.8 Million by 2032

content: Allied Market Research reveals that the global carbon credit trading platform market, valued at \$112.4 million in 2022, is set for remarkable growth. It's projected to reach \$556.8 million by 2032, growing at a CAGR of 17.4% from 2023 to 2032. A carbon credit trading platform serves as an online marketplace enabling the buying and selling of carbon credits, which are rights to emit specific amounts of greenhouse gases. This plays a crucial role in

helping businesses, governments, and organizations meet emissions reduction goals and contribute to climate change mitigation.

doc_url: [Masked URL 2]

pub_code: [Masked pub_code 1]

source: [Masked source 1]

section: [Masked section 1]

author: [Masked author 1]

region: American